physics, the first of these propositions is completely meaningless (not even wrong), and the second one means something quite different from what you might think. It all comes down to a deep logical difference between the classical and quantum concepts of the state of a system. Explaining the quantum concept of state will require some abstract mathematics, so let's pause for a brief interlude on complex numbers and vector spaces. The need for complex quantities will become clear later on, when we study the mathematical representation of spin states.

1.8 MATHEMATICAL INTERLUDE: COMPLEX NUMBERS

Everyone who has gotten this far in the Theoretical Minimum series knows about complex numbers. Nevertheless, I will spend a few lines reminding you of the essentials. Fig. 1.6 shows some of their basic elements.

A complex number z is the sum of a real number and an imaginary number. We can write it as

$$z = x + iy,$$

